



**Faculty of Engineering & Technology Electrical & Computer Engineering
Department**

Circuits and Electronics Laboratory - ENEE2103

PreLab Exp4

Prepared By:

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Instructor: Dr. Nasser Ismail.

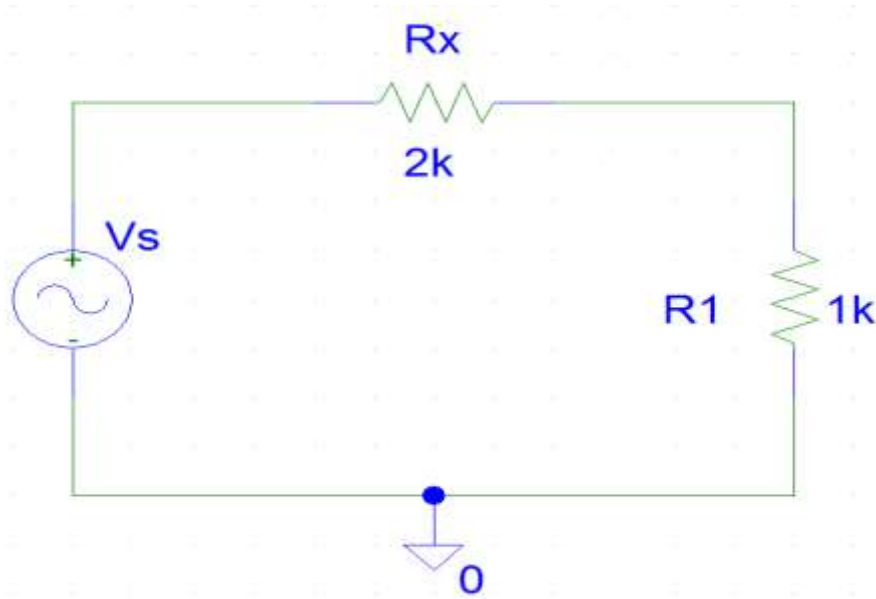
Assistant: Eng. Motasem

Section: 3

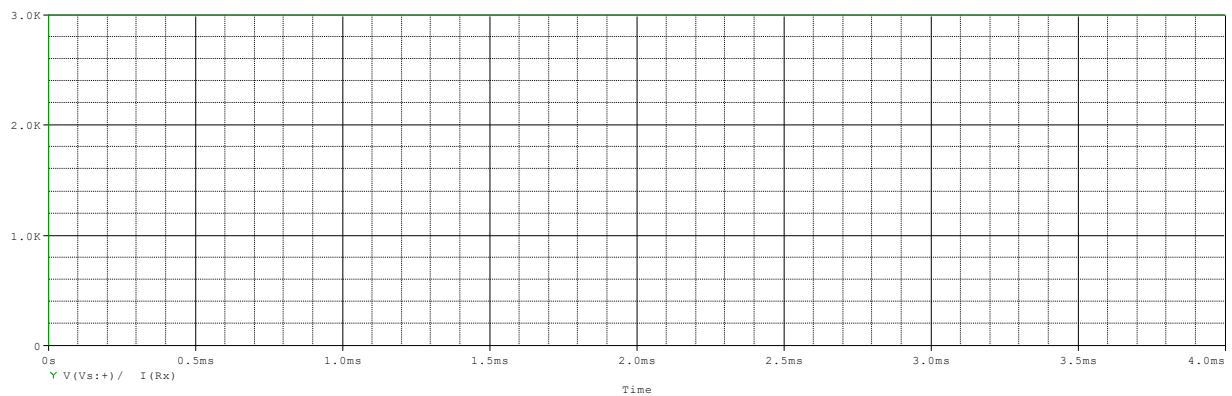
Date: 5/8/2023

Part A:

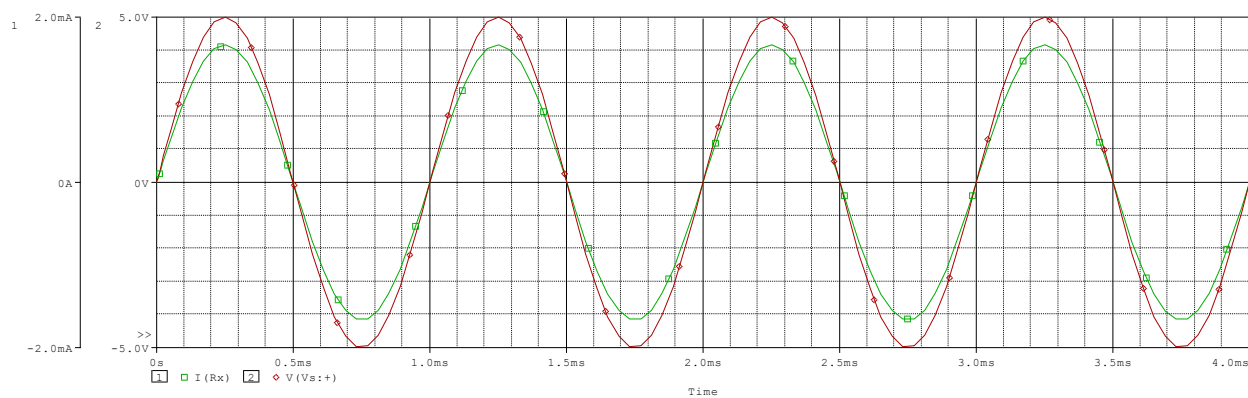
- For Fig 4.1:



- For freq = 1Khz:

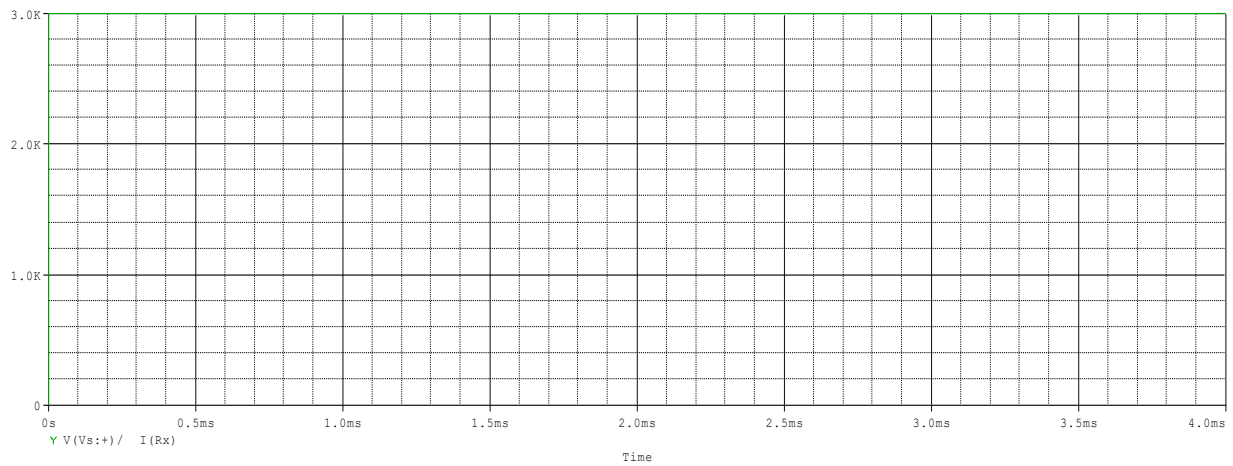


Total Impedance = 3K-Ohm

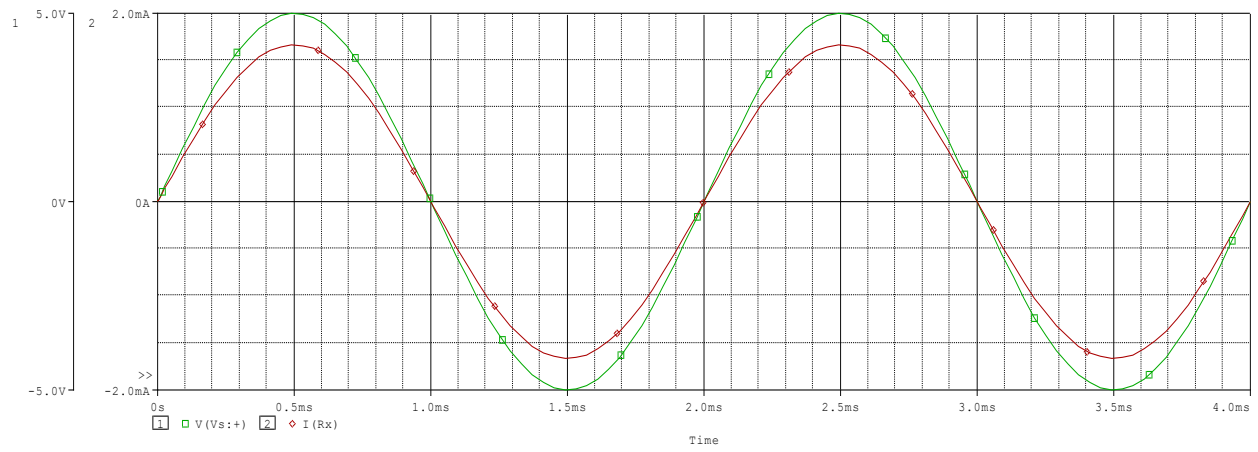


Phase Shift between V & I = 0 (V & I are inphase).

➤ For Freq = 0.5Khz:

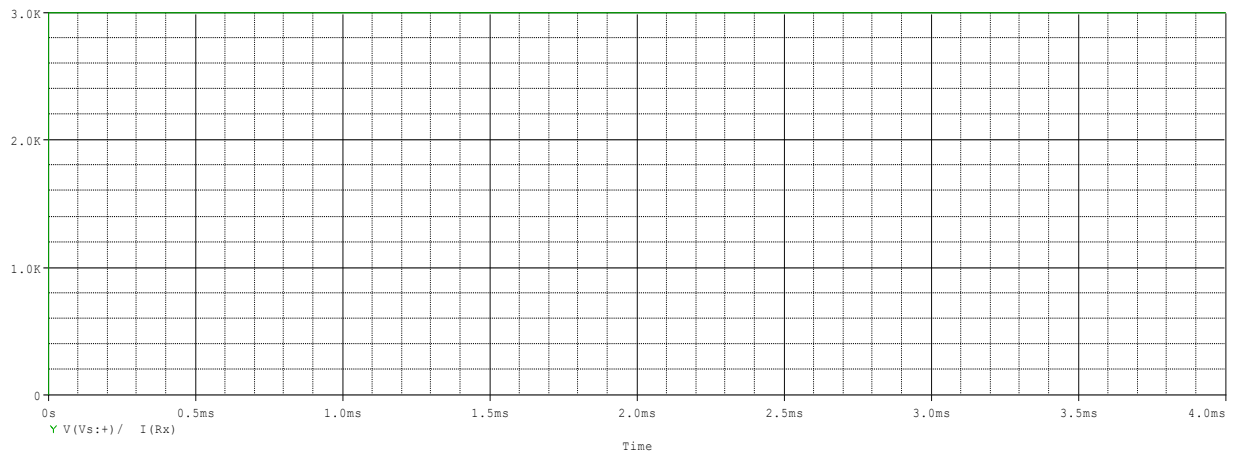


Total Impedance = 3K-Ohm

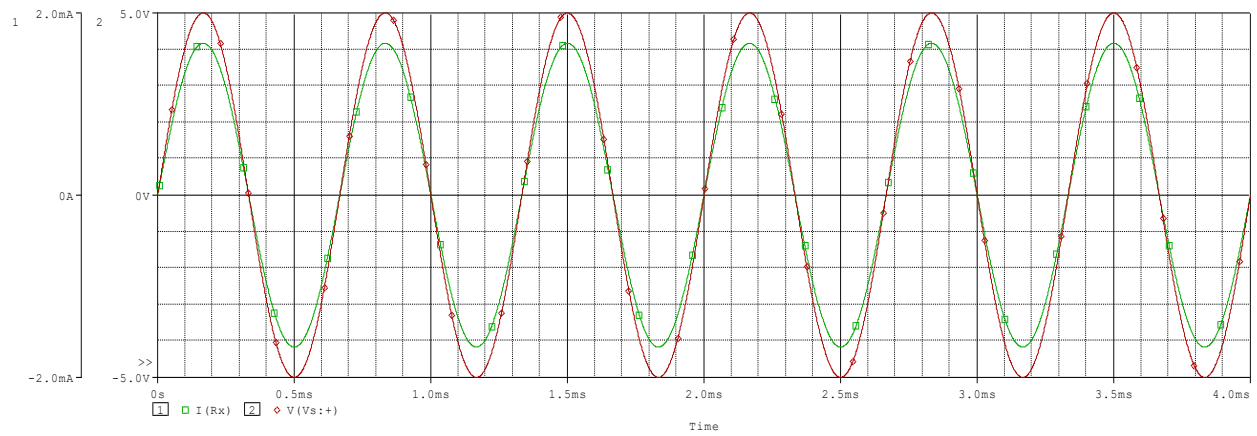


Phase Shift between V & I = 0 (V & I are inphase).

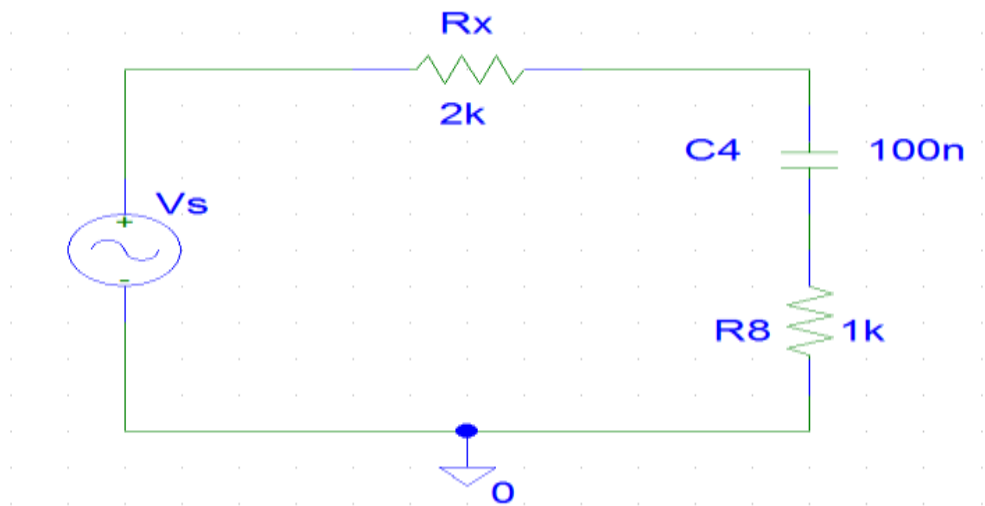
➤ For freq = 1.5KHz:



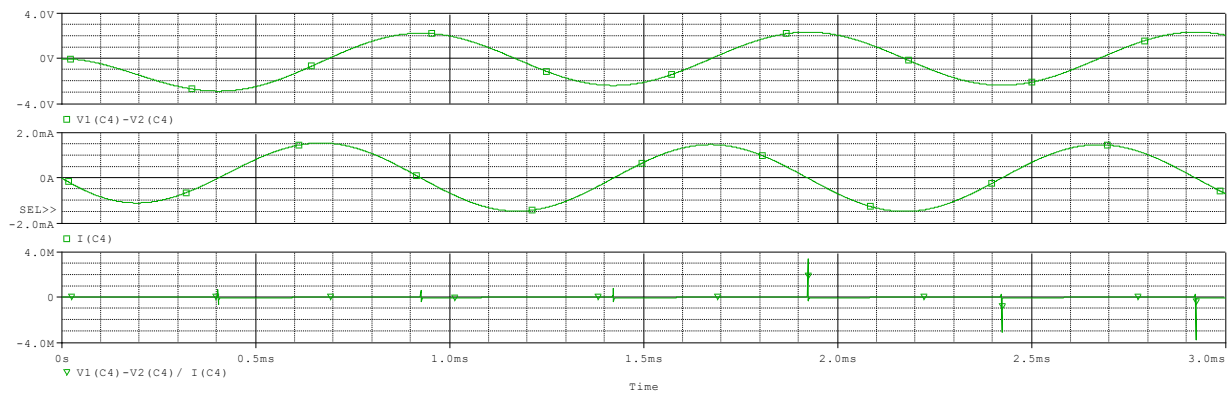
Total Impedance = 3K-Ohm



- For fig 4.2:

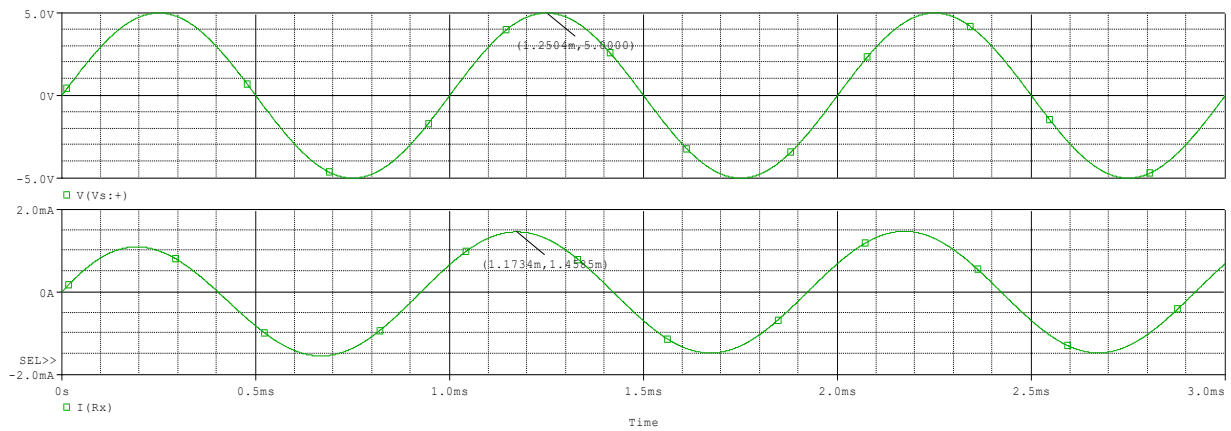


- For freq = 1KHz:



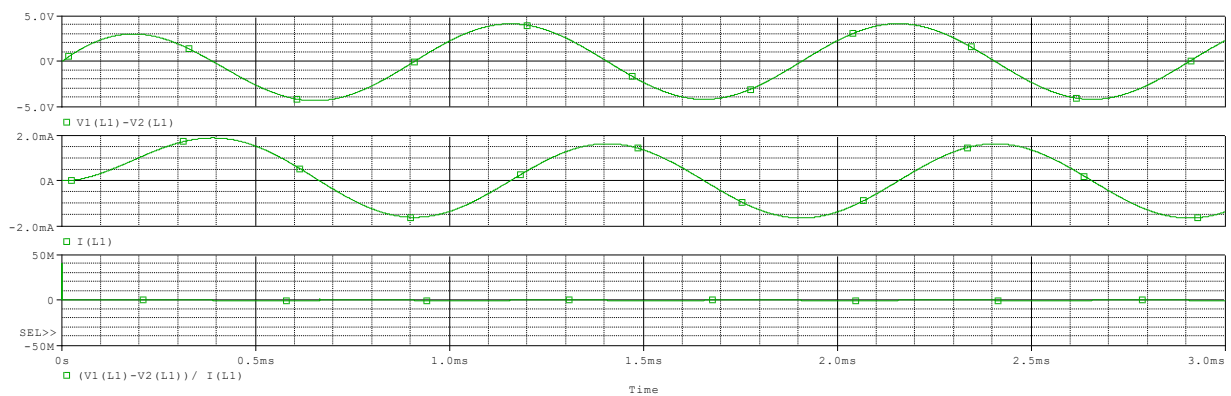
Total Impedance = $3k - j 1.53K$ (Ohm)

Total Impedance (Magnitude) = 3.37 K-Ohm



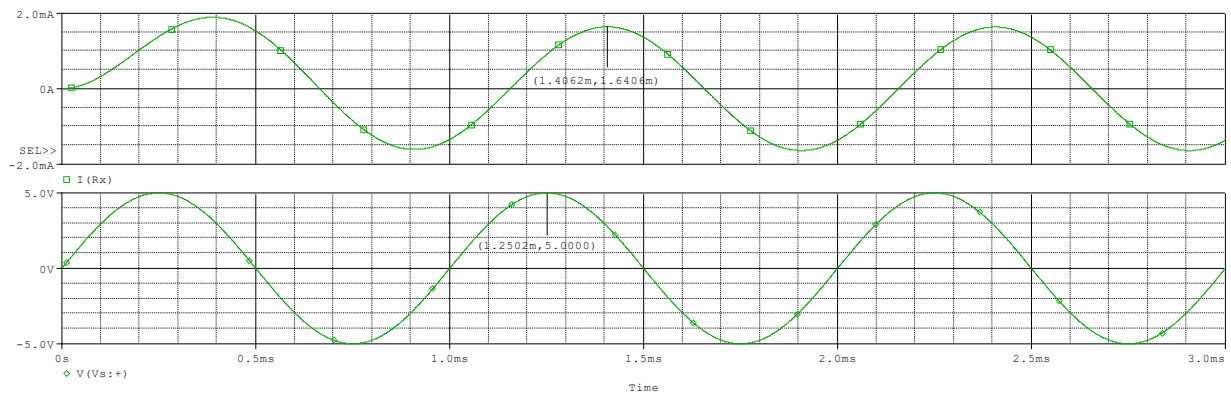
Phase shift = $2 \cdot \pi \cdot f \cdot (\Delta t) = 360 \cdot 1000 \cdot (77\mu) = 27.72$ degree

- For fig 4.3:
- For Freq = 1KHz:



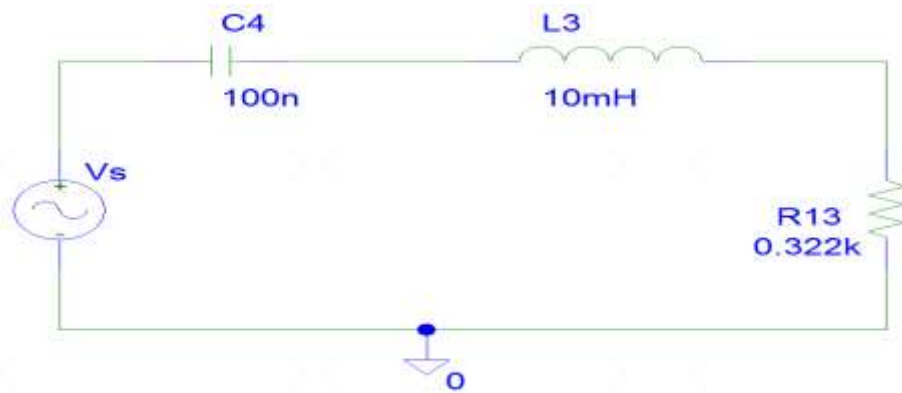
Total Impedance = $3k + j 2.54K$ (Ohm)

Total Impedance (Magnitude) = 3.93K-Ohm

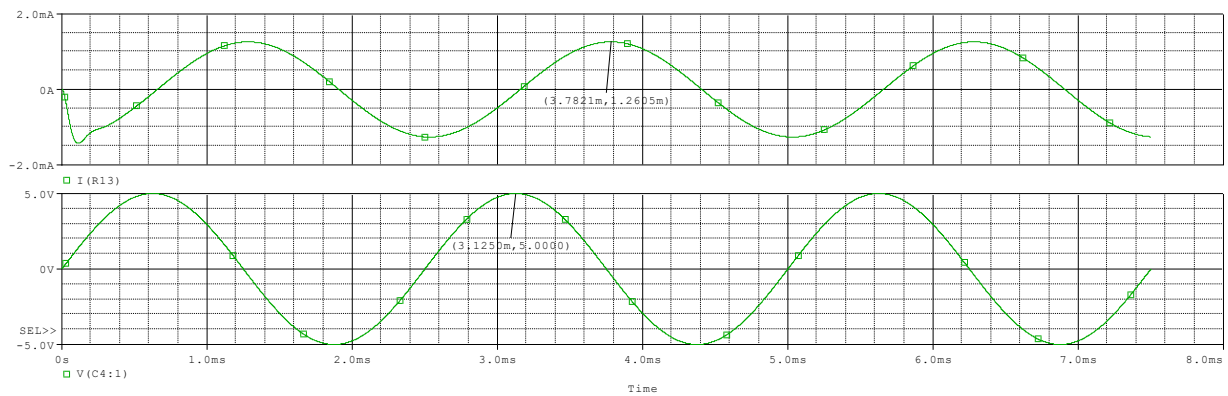


Phase shift = $2 \cdot \pi \cdot f \cdot (\Delta t) = 360 \cdot 1000 \cdot (156\mu) = 56.16$ degree

Part B:

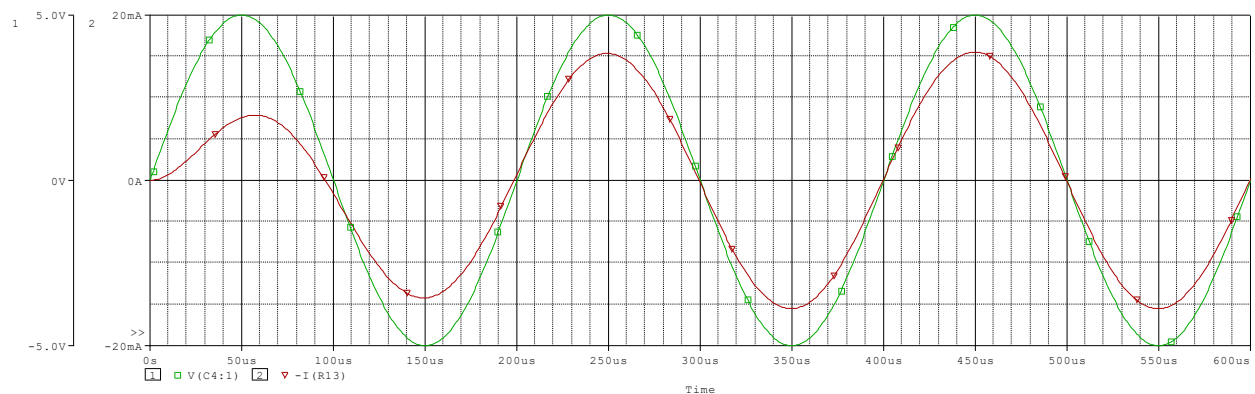


- For Fig 4.4:
- For Freq = 400Hz:



Phase shift = $2\pi f \cdot (\Delta t) = 360 \cdot 400 \cdot (0.66\text{ms}) = 95^\circ$

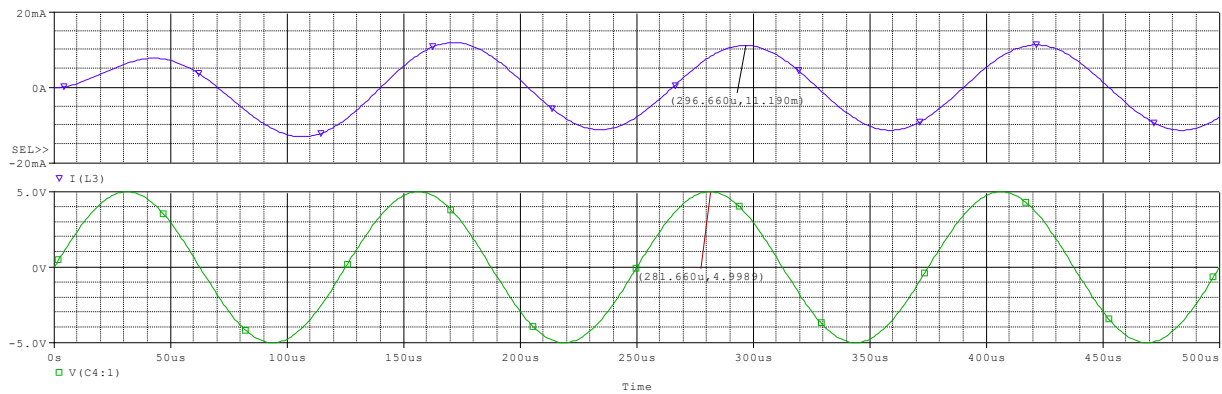
- For Freq = 5KHz:



Phase Shift = 0 (they are inphase)

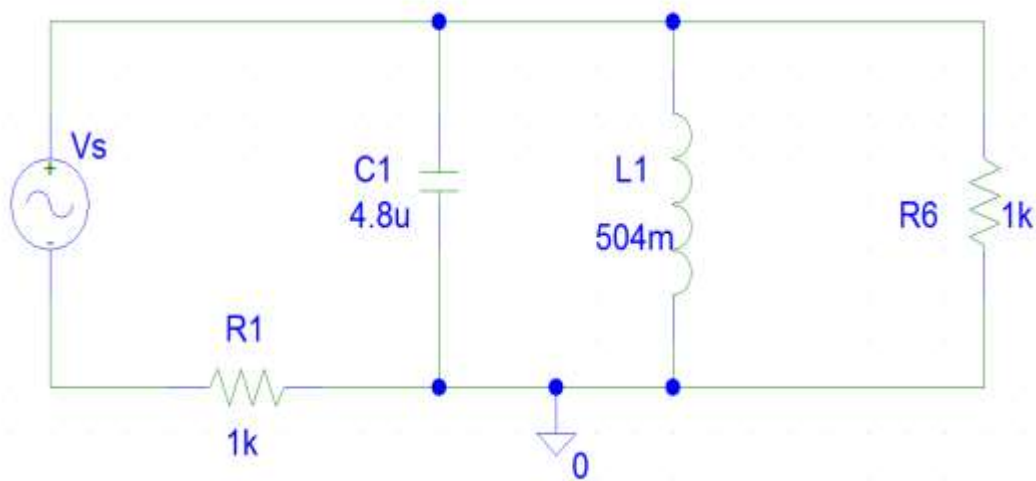
So, $f = 5\text{KHz}$ is the resonance freq.

➤ For Freq = 8KHz:

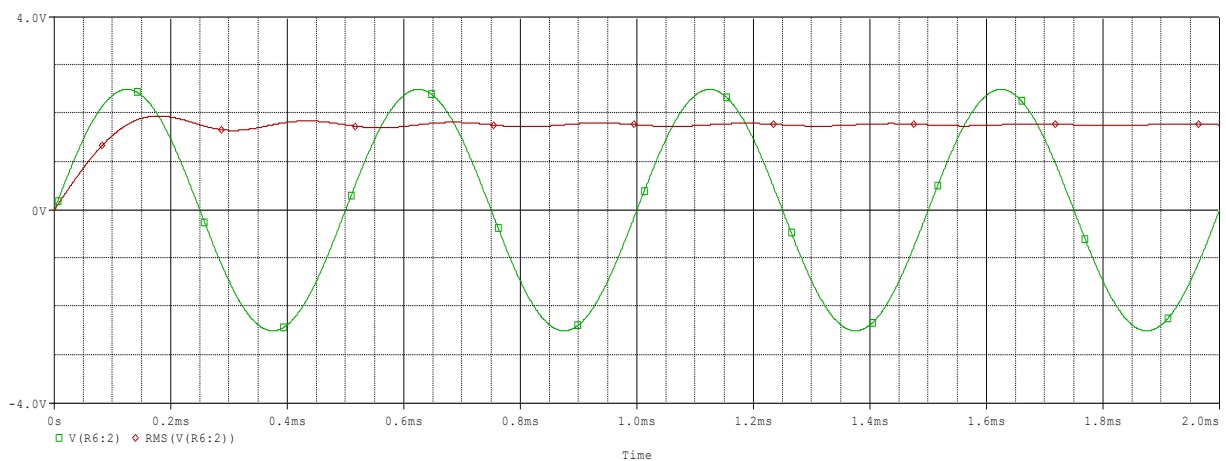


$$\text{Phase shift} = 2\pi f (\Delta t) = 360 \times 8K \times (0.66m) = 43.2 \text{ degree}$$

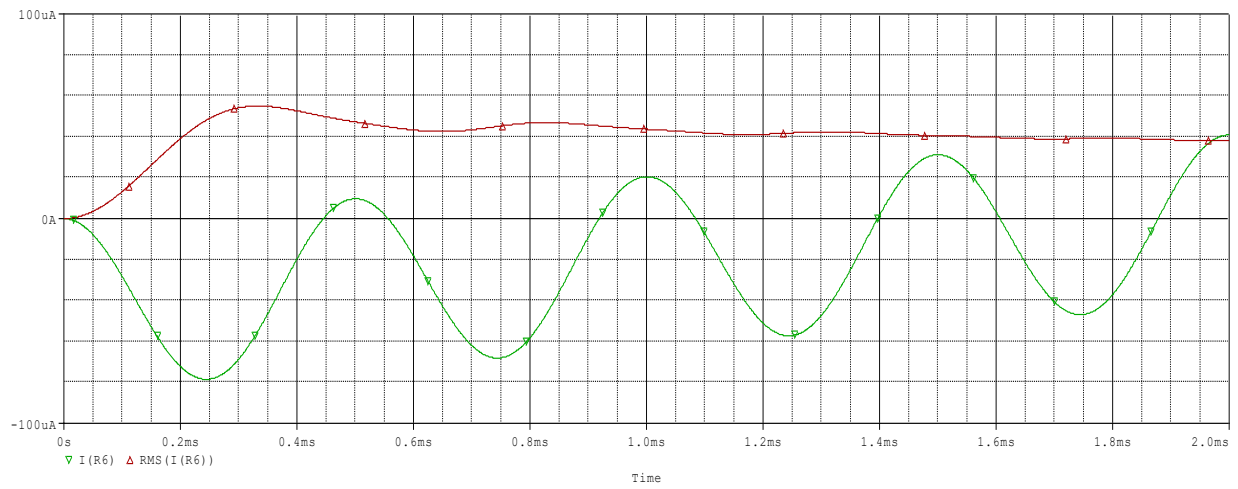
Part C:



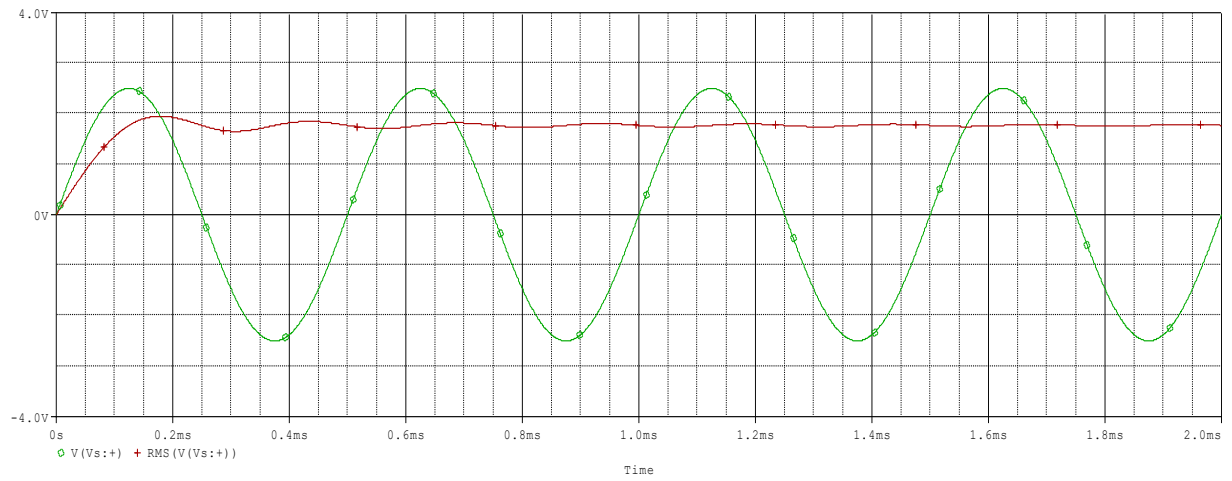
➤ RMS Voltage Across R6:



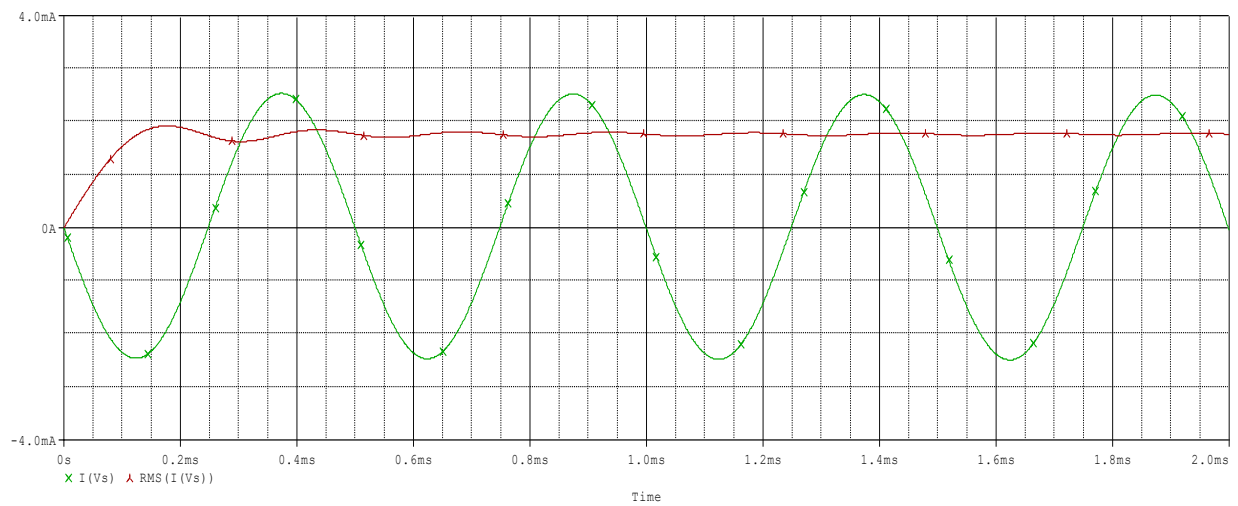
➤ RMS Current Across R6:



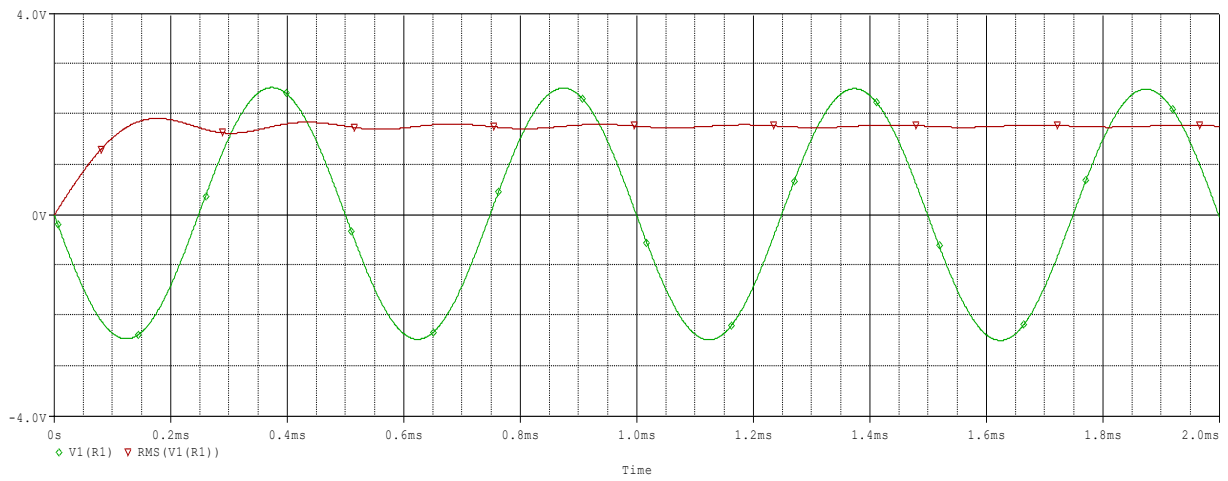
➤ RMS Voltage for Vs:



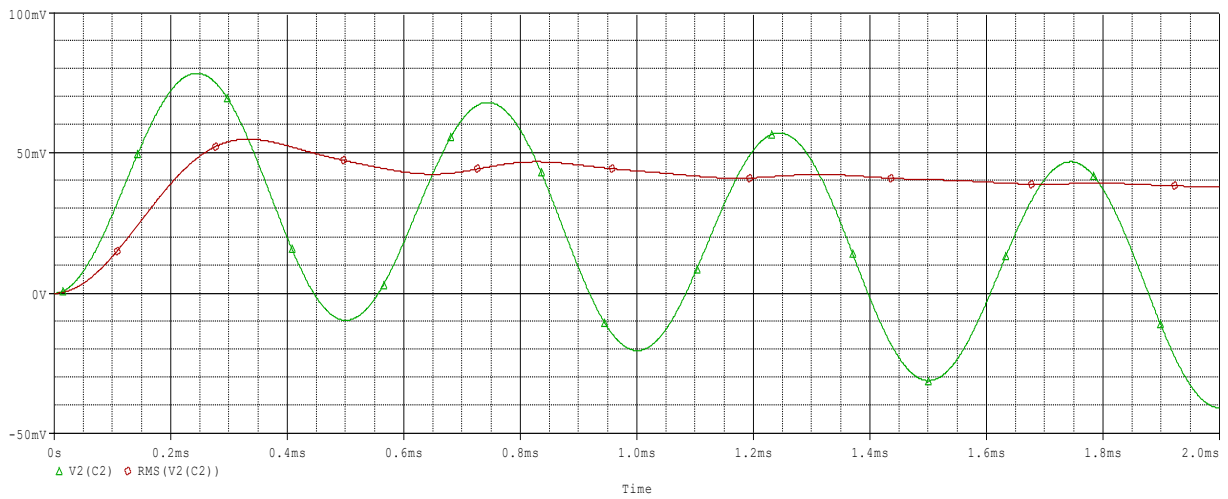
➤ RMS Current for Is:



➤ RMS Voltage Across R1:



➤ RMS Voltage Across C2:



➤ RMS Current Across C2:

